

# B04 Quick Revision

Warehouse boundaries, history, quality and recovery

## Closed-book revision map

| Lesson  | Recall contract                                                                                                    |
|---------|--------------------------------------------------------------------------------------------------------------------|
| B04-L01 | Separate OLTP event recording from OLAP historical decision workloads; protect the operational source.             |
| B04-L02 | Choose warehouse/lake/lakehouse from data shape, governance, consumer and replay needs, not product branding.      |
| B04-L03 | Separate business identity from warehouse version identity; SCD history cannot have ambiguous effective intervals. |
| B04-L04 | Name durable landing and transformation boundary; raw evidence must remain replayable.                             |
| B04-L05 | Layer contracts define allowed state, transformations, consumers and promotion gates.                              |
| B04-L06 | Choose batch/micro-batch/streaming from SLA, late-data and replay trade-offs.                                      |
| B04-L07 | Lineage + metadata + ownership must make impact and freshness explainable.                                         |
| B04-L08 | Quality rules need severity, owner and block/warn/reconcile action.                                                |
| B04-L09 | Replay/backfill must be bounded, idempotent where required and must not mutate raw evidence.                       |

## Failure-first recall

- Inventory and other snapshot measures are semi-additive across time: summing daily snapshots can create a believable but meaningless total.
- A current-master row is not a historical truth table. Same-day conflicting changes are a blocker until resolved.
- Orphan records must remain visible as quality evidence; deleting them to make joins clean is a critical failure.
- Many-to-many tags/memberships can multiply transaction measures if joined before aggregation.
- Higher-grain targets belong at their own grain; compare only after actuals are aggregated compatibly.
- Replay is safe only when the source slice, watermark/window, transformation version, destination behavior and reconciliation are explicit.

## Move on when

With the book closed, you can classify workload/storage/layers, design history, state latency and lineage contracts, define block/warn quality rules, and explain a bounded replay plan with independent reconciliation.